Dimas Shidqi Parikesit

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Research focus: Formal methods, cloud and distributed systems, systems correctness.

Education

University of Virginia Aug. 2024–Present
PhD in Computer Science (in progress) | Advisor: Prof. Chang Lou

Bandung Institute of Technology, Indonesia Aug. 2020–Aug. 2024
BSc in Informatics | GPA: 3.80/4.00

Experiences

Research Assistant - Software Correctness Ongoing research
LiftLab, University of Virginia

- Investigate how to effectively use **LLM** to carry **useful semantics information** from original systems into reimplementations with entirely different designs or architectures.
- Reviewed **12 software-reimplementation cases**, including the SGLang router, Bun, and Vinext.

Research Assistant - Distributed-System Reliability (LISA) Aug. 2024–Present
LiftLab, University of Virginia

- Lead system design and implementation of LISA, using **LLM to prevent regression failures** by turning historical bug reports into **low-level invariants**.
- Identified **34 new bugs** across ZooKeeper, HDFS, HBase, and Solr; 16 were confirmed by developers. Examples: missing block locations in HDFS; expired snapshots accepted by HBase.
- Modified the JVM engine and built the full multi-stage pipeline to check inferred invariants using **static analysis, concolic execution, and Z3**.
- Lead and mentor **five research interns**; delegate case collection and evaluation, and guide research practices.

Undergraduate Researcher - Runtime Failure Detection (T2C) Nov. 2022–Dec. 2023
OrderLab, University of Michigan | Collaboration with Microsoft Research

- Conducted the full evaluation of T2C, which derives **runtime semantic checkers from tests**, across ZooKeeper, Cassandra, HDFS, and HBase; detected **15 of 20 reproduced real-world silent failures**.
- Implemented a **trie-based representation** of assertion operation sequences, sharing common prefixes instead of storing separate operation lists to reduce initial runtime overhead by more than 50%.

Undergraduate Researcher - Inference Caching (EVProx) Apr. 2022–Jun. 2022
UCare, University of Chicago

- Built **EVProx cache-key computation** to generate approximate values as L3 cache layer to reduce 99th tail latency of Facebook Deep Learning Recommendation Model (DLRM) by 22%; Ran experiment on **NVIDIA A6000**

Software Engineer Intern May. 2022–Jul. 2022
Capital Dynamics, Sdn. Bhd.

- Rebuilt 6 company websites using modular front end components and shared CMS.

Selected Publications

ACM Hot Topics in Networks (HotNets), 2025 | First author; position paper
(LISA) Once Bitten, Still Shy: Can We Prevent Cloud Systems from Repeating Their Mistakes?

USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2025 | Second author
(T2C) Deriving Semantic Checkers from Tests to Detect Silent Failures in Production Distributed Systems

Technical Skills

Languages: Java, Python, Rust, C++, Bash.
Systems: Linux, JVM, ZooKeeper, Hadoop/HDFS, HBase, Cassandra, Solr.
Methods: Formal methods, SMT Solver (Z3), Static analysis, concolic execution, runtime checking.
AI: PyTorch, vLLM, SGLang, NVIDIA Dynamo, Ray.

Honors, Awards & Research Support

Microsoft Accelerating Foundation Models Research Initiative (US$15,000 in Azure API credits) 2025
SOSP 2024 Student Travel Grant 2024
First Place, Datathon AI, Indonesia 2021
Gold Medal, National Science Olympiad, Indonesia 2016